Guide for Establishing Utility-Based Computing

Overview with a phased build out plan

Goals for Technical Architecture

- Build a high availability IT environment capable of handling planned and unplanned threats to availability.
- Build a resilient infrastructure that provides uninterrupted service in the event of man-made or natural events that stress the environment, systems, processes and people.
- Build a portfolio of technology services that are made globally available to meet the needs of the business.
- Build an adaptable technology platform that can easily accommodate changes (from both the business and technology).

Key Definitions

- High Availability means…
  - Going beyond compliance
  - Adopting a cradle-to-cradle approach
  - Implement precautionary principles
  - Acknowledge the importance of risk factors and setting clear compliance goals
- Resilient means…
  - Resistant to unplanned disruptions.
  - Security-rich environment that help mitigate risk.
  - Transparently adaptive to change.
  - Scalable to upswings and downswings in the market.
  - Assists in rapid recovery should a failure or disruption occur.

Guiding Principles for Technical Architecture

- **Strive for Simplification**
  - Reducing complexity will reduce costs and will help to reduce risk.
  - Tools should be evaluated for their ability to address our features while helping us reduce the complexity of current technical architecture.
  - Review the platforms, products and services in the Technical Architecture and look for opportunities to consolidate, retire and replace.
  - Tools should be chosen for their ability to improve quality, reduce cycle time and reduce TCO.
  - Simplify policies and SOPs where possible. Automate policies using business rules to drive processes (services where automation permits).

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Guiding Principles for Technical Architecture

- **Ubiquitous Deployment**
  - Products and services need to be globally available. As the rate of technological change accelerates, it is imperative that our solutions to the business and end users via ubiquitous methods.
  - We should avoid proprietary solutions.
  - Utilize a common infrastructure (i.e. The Internet), so we can avoid the need for custom ($$$) solutions.
    - Example: Use SSL and SSH rather than setting up VPNs to reduce the potentially costly setup and ongoing maintenance costs of VPNs. This can be achieved simply by utilizing existing web browsers; T1s and firewalls without special configuration/maintenance issues.

Guiding Principles for TA

- **Adapt to a High Rate of Change**
  - The ability to not just tolerate change but embrace change in its very design – allowing the business and end users to adapt to future changes as a matter of course rather than exception.

- **Manage Technology as an Investment**
  - View technology as an investment, rather than an annual expense, that will yield a return in exchange for up-front expenditures and assumption of risk.

Guiding Principles for Technical Architecture

- **Strategic Commitment to the Compliance Mgmt**
  - Going beyond compliance
  - Adopting a cradle-to-cradle approach
  - Implement precautionary principles
  - Acknowledge the importance of risk factors and setting clear compliance goals

- **Customer Focused Delivery**
  - Measure and assess people, projects and teams by what they accomplish on behalf of the customer. Metrics should reflect impact on Quality, Productivity or Cost as it relates to the services we provide for customers.
  - Positive Trending is Good !

Guiding Principles for Technical Architecture

- **Don't Be Beholden to a Single Vendor FOREVER**
  - While single sourcing might drive down costs, single source solutions may preclude Hospira from leveraging new technologies to further contain cost or increase agility
  - Products should be evaluated for their Fit, Form & Function (3Fs) as well as their ability to positively impact the customer (Quality, Productivity, Costs).

- **Enhance Individual Employee Productivity**
  - Provide flexible and scalable tools that individuals can use to perform their roles more effectively. Note: Assume significant workload growth and complexity over time

Creating a "Bridge" to the future

This is where we need to make the logical leap from what we have today, what we are trying to do today, what the reason is as to how we got to today and then provide a logical transition to Utility Computing

The Future Direction is Utility Computing

What is "utility computing" ?

On Demand
Peak Load
Capability

Ubiquitous

Highly
Available

High availability is a balance between the need to support the business requirements and goals at an affordable cost.

Pay (QoS)
Services Levels

Location
Agnostic
(on or off premise)

Scope of Technical Architecture

In order to achieve our goals for highly available services, we need to address resiliency throughout all the architecture disciplines (Business, Information and Technology). Encompasses more than just the Technical Architecture…

Business Architecture:
Strategy
Organization
Process (IT & Business)

Information & Application Architectures:
Business Logic & Rules
Applications & Tools
Service Architecture
Data Architecture

Technology Architecture:
Presentation Layer (Portal)
Security Framework
Hardware Platforms
Physical Environment

Benefits of Adopting Utility Computing

- Enables organizations to carry out an infrastructure transformation effort that is manageable, cost effective and minimally disruptive to ongoing business operations.

- Allows IT to quickly take control of technology-related costs, enhance system security and improve our ability to support business strategies.

- Provides a way for IT to self-fund longer-term, technology - enabled improvements that will drive additional value

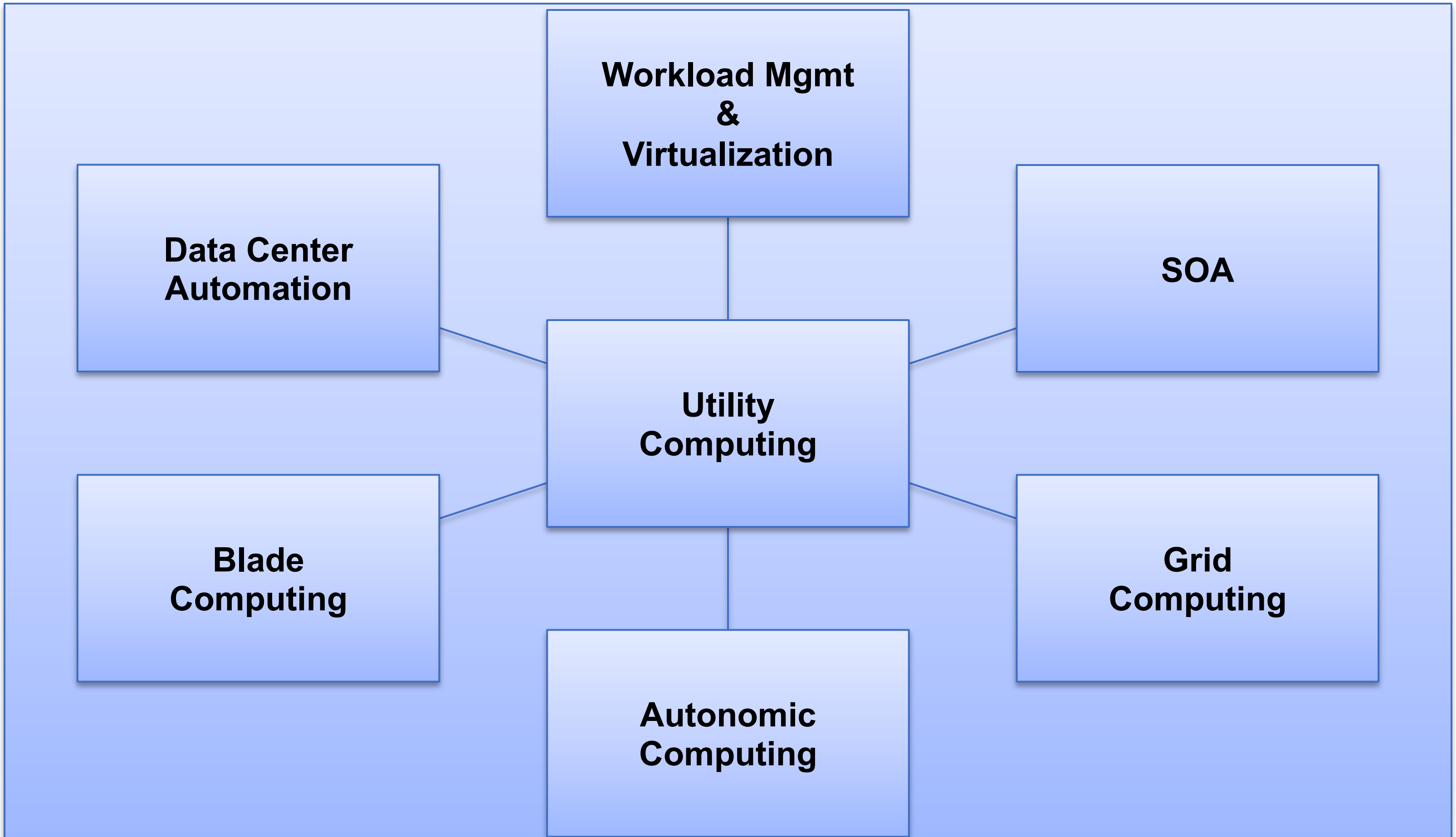
Future State for the Technical Architecture

- Move towards a utility computing model, which will allow us to anticipate and adjust to both planned and unplanned events.

- This model will leverage a number of technologies and best practices to provide us with prudence and foresight, coupled with a flexible and adaptable capability for managing risk (compliance, business continuity, security for all assets including intellectual property) while enabling the business to easily change to meet future market demands. (wordy and unclear)

- The future state of the TA will serve as the foundation for meeting an emerging standard in business resilience for pharmaceutical and life sciences. (What emerging standard?)

Future State (Conceptual View)



Future State (Details)

- **Workload Management & Virtualization**
  - These products provide infrastructure automation and workload management for complex, heterogeneous environments by leveraging input from the IT infrastructure to "sense" changes and make real-time decisions about where and when resources need to be deployed.
  - These tools use provisioning to dynamically allocate and configure infrastructure capacity, from servers to storage to networking, to meet changing business needs.
  - The result is a data center that can react dynamically to change, enabling the business to become more flexible, scalable and less complex while at the same time reducing costs.

Future State (Details)

- **Workload Management & Virtualization**
  - Application virtualization technology will allow organizations to move an application from one server to another without disruption and provide usage-based metering to enable organizations to measure and monitor total usage of computing resources.
  - Citrix and other thin-client software technologies are enabling workers to tap into a utility-style IT environment by providing rapid, secure deployment of full-featured applications from a central location to users on any device or connection.

Future State (Details)

- **Service Oriented Architecture**
  - Services will allow utility computing providers to provide specific processes and application services without regard to the technology that the customer uses
- **Data Center Automation**
  - Intelligent software that will enable the provisioning of resources, configuration management and metering

Future State (Details)

- **Grid Computing**
  - Grid computing utilizes distributed computing resources to create a virtual computing platform. This will be leveraged to create a highly scalable and virtualized data center infrastructure
- **Blade Computing**
  - Blade-based systems will facilitate horizontal application scalability via modular and compact servers, ease of deployment and delivery of a high degree of performance

Future State (Details)

Autonomic Computing

Will play an important role in data center automation (range of technologies to enable systems to heal automatically and to prevent failures)

Self-configuring
Adapt automatically to the dynamically changing environments

Self-healing
Discover, diagnose, and react to disruptions

Self-optimizing
Monitor and tune resources automatically

Self-protecting
Anticipate, detect, identify, and protect against attacks from anywhere

ROADMAP

Phased approach for adopting and implementing a technical architecture based on utility computing

Phase 1

- **Infrastructure stabilization & optimization**
  - Objective: Establish a dynamic and scalable utility computing-style infrastructure within the firewalls of the enterprise.
  - Key Activities:
    - standardizing technologies onto fewer platforms
    - eliminating outdated applications (and consolidating the remaining ones onto fewer servers)
    - consolidating storage devices
    - aligning the performance of infrastructure devices and application portfolios to business goals
    - exploring the opportunities to outsource some or all IT systems and services to third-party vendors.

Phase 2

- **Infrastructure Virtualization & IT Delivery**
  - Objective: Establish an on-demand capability by aligning IT and business (including manufacturing) processes with customer demands..
  - Key Activities:
    - Move applications among various processing resources (including servers and storage) within their data centers to optimize performance across the enterprise. This also allows organizations to shave off the peak load costs associated with running applications.
    - Allocate capacity and resources—such as utility-based data centers, mobile work scenarios, workload management and IP (voice and data) services—dynamically and automatically.
    - More easily identify—and manage—the resources dedicated to specific IT functions.

Phase 2 (Cont.)

❑ Infrastructure Virtualization & IT Delivery

- Objective: Establish an on-demand capability by aligning IT and business (including manufacturing) processes with customer demands..
- Key Activities:
  - Reduce the complexity of managing hardware from multiple vendors.
  - Implement a simplified interface between IT resources and business processes by implementing a portal strategy.
  - Modularize services to businesses at various levels, including storage, computing, and databases.
  - Measure provisioning time for new applications in seconds (not days) and response times for change requests in minutes.
  - Eliminate "downtime" due to reduced hardware maintenance..

Phase 2 (Cont.)

❑ Infrastructure Virtualization & IT Delivery

- Objective: Establish an on-demand capability by aligning IT and business (including manufacturing) processes with customer demands..

- Key Activities:
  - Establish a "pay for use" strategy with our vendors for computing resources
  - Virtualize the computing pool so we can run diverse operating systems, such as UNIX, Windows and Linux. (leverage a combination of enabling technologies along with virtualization (such as logical partitioning, workload management and dynamic server clustering)
  - Move towards "self-healing," which means they incorporate technologies that allow them to anticipate, detect and protect themselves from attacks and self-diagnose and self-configure whenever disruptions arise.

Phase 2 (Cont.)

❑ Infrastructure Virtualization & IT Delivery

- Objective: Establish an on-demand capability by aligning IT and business (including manufacturing) processes with customer demands..

- Key Activities:
  - Virtualize the storage pool by eliminating Direct-attached storage (DAS) and migrate all applications to SANs.
  - Virtualize the network to link everyone to everything, everywhere by implementing Virtual Private Network (VPN) solutions.
  - Implement higher-quality, lower-cost IP bandwidth and quality-of-service tools, which allow service providers to deliver differentiated service and enterprises to prioritize traffic based on application, user or location.
  - Converge voice and data infrastructures to reduce telecommunications support and administrative costs.

Phase 2 (Cont.)

❑ Infrastructure Virtualization & IT Delivery

- Objective: Establish an on-demand capability by aligning IT and business (including manufacturing) processes with customer demands.

- Key Activities:
  - Leverage wireless technologies, which help fuel "anytime, anywhere“ connectivity and the development of new applications.
  - Implement optical networking solutions and high-bandwidth Local Area Network (LAN) and Metro Area Network (MAN) technologies, to drive large capacity through low-cost bandwidth.
  - Migrate towards tools that centrally manage our routers, bridges and hubs, and re-deploy resources through a VLAN configuration.
  - Implement load balancers and establishing a firewall pool
  - Implement Identity management technology

Phase 2 (Cont.)

❑ Infrastructure Virtualization & IT Delivery

- Objective: Establish an on-demand capability by aligning IT and business (including manufacturing) processes with customer demands.

- Key Activities:
  - Build an integrated security management framework to simplify the cost of maintaining a dynamic security infrastructure. Today firewalls, intrusion detection systems, anti-virus agents, web access control systems, etc. are all independently-managed technologies. Management integration enhances security by centralizing the monitoring of events, which reduces operational costs and improves efficiency. The deployment of each distributed agent and its configuration is managed via a single point of control

Phase 3

- **Leveraging the Dynamically Provisioned TA**
  - Objective: Extend the on-demand capability beyond the enterprise to 3rd party resource providers. This is the realization of virtual services with unlimited scale and capability.
  - Key Activities:
    - Enable clients to tap into IT capacity from 3rd party clients anywhere while leveraging a pay-as-you-go model for consumption.
    - Implement new resource management models, new training programs for managers and staff.
    - Develop new governance models to handle internal and external SLAs for on-demand services.

Thank You

Measure what matters
Create and deliver the right value
Build and nurture a winning culture
Proactively manage change & complexity

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